THE AUTOMORPHISM GROUP OF A NORMAL EXTENSION ACTS TRANSITIVELY ON THE MAXIMAL IDEALS OF THE INTEGRAL CLOSURE

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file more-algebra.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The chapter file pinned-original-source.tex carries only \input{preamble} on its first line, so no preamble text lives inside the chapter. The retained file sources/stacks-more-algebra/retained-preamble.tex is a mechanical, line-for-line copy of the project's single shared preamble as staged in the snapshot; nothing was added, removed or redefined and no package was added.

Editorial note. The source URL is the source-level Stacks Project URL, as required when more than one card is produced from one source; the per-claim tag is carried by claim_key.

Editorial note. The single \ref pointer inside the counted proof span refers to the construction remark and is recorded as a non-load-bearing navigation pointer.

Allowed standard prerequisites (editorial).

- The rings K_1, L_1, A_1 and B_1 attached to an extension of discrete valuation rings and a normal extension. the construction invoked at statement lines 34612-34614. Fixes the objects on which G acts and among whose maximal ideals transitivity is asserted.
- The reduced tensor product (L tensor_K K_1)_red and the canonical map from it in characteristic p by a large p-power. used at proof lines 34623-34630. Is the device that lifts the G-invariant element b' back to L tensor_K K_1.
- Normality of the discrete valuation ring B and the identity K = (K_1)^G for a normal extension. used at proof lines 34631-34632. Forces the lifted invariant element to lie in B, which is what produces the contradiction.

Source reference. Original remark 33882: sources/stacks-more-algebra/pinned-original-source.tex, source lines 33882–33882. Same-chapter navigation pointer; the statement it names is standard for the section and the argument that consumes it stays inside the counted proof span.

Lemma 0.2. *Let $A \to B$ be an extension of discrete valuation rings with fraction fields $K \subset L$. Let K_1/K be a normal extension. Say $G = Aut(K_1/K)$. Then G acts on the rings K_1, L_1, A_1 and B_1 of Remark 33882 and acts transitively on the set of maximal ideals of B_1.*

Proof. Everything is clear apart from the last assertion. If there are two or more orbits of the action, then we can find an element $b \in B_1$ which vanishes at all the maximal ideals of one orbit and has residue 1 at all the maximal ideals in another orbit. Then $b' = \prod_{\sigma \in G} \sigma(b)$ is a G-invariant element of $B_1 \subset L_1 = (L \otimes_K K_1)_{red}$

Date: stacks-project:more-algebra.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

which is in some maximal ideals of B_1 but not in all maximal ideals of B_1. Lifting it to an element of $L \otimes_K K_1$ and raising to a high power we obtain a G-invariant element b'' of $L \otimes_K K_1$ mapping to $(b')^N$ for some $N > 0$; in fact, we only need to do this in case the characteristic is $p > 0$ and in this case raising to a suitably large p-power q defines a canonical map $(L \otimes_K K_1)_{red} \to L \otimes_K K_1$. Since $K = (K_1)^G$ we conclude that $b'' \in L$. Since b'' maps to an element of B_1 we see that $b'' \in B$ (as B is normal). Then on the one hand it must be true that $b'' \in \mathfrak{m}_B$ as b' is in some maximal ideal of B_1 and on the other hand it must be true that $b'' \notin \mathfrak{m}_B$ as b' is not in all maximal ideals of B_1. This contradiction finishes the proof of the lemma. $\square$